

ZHIQI (BIRKHOFF) CHENG

info@birkhoff.ch | github.com/birkhoffcheng | linkedin.com/in/birkhoffcheng

EDUCATION

University of Illinois Urbana-Champaign

August 2019 – May 2023

Bachelor of Science in Computer Science

GPA: 3.94 / 4.00

- CS 374 - Algorithms & Models of Computation
- CS 418 - Interactive Computer Graphics
- CS 423 - Operating Systems Design
- CS 424 - Real-time Systems
- CS 425 - Distributed Systems
- CS 426 - Compiler Construction
- CS 433 - Computer System Organization
- CS 463 - Computer Security II
- CS 498 - Applied Cryptography

PROFESSIONAL EXPERIENCE

Jump Trading

Chicago, IL

Technical Operations Intern

June 2022 — August 2022

- Proposed, implemented, and tested solutions to allow molecule tests to run podman in Kubernetes cluster
- Improved various Ansible roles to accelerate the development, testing, and deployment of internal applications

Rivos Inc.

Mountain View, CA

Software Engineer Intern

January 2022 — May 2022

- Added QEMU support for RISC-V CFI extension with shadow stack and landing pad instructions
- Enabled Linux kernel to allow user programs to control RISC-V CFI status at run time
- Implemented alternative floating point format and native float support in Rivos accelerator simulator
- Designed a better test interface for the accelerator simulator to allow per-instruction testing

National Center for Supercomputing Applications

Urbana, IL

Advanced Cyberinfrastructure Intern

August 2021 — December 2021

- Assisted in installation and security maintenance of supercomputer clusters at NCSA
- Developed a Linux security hardening guideline and scripts for setup and enforcement
- Integrated NCSA's security requirements into a Lynis plugin to automate security vetting

University of Illinois Urbana-Champaign

Champaign, IL

Course Assistant

January 2020 — December 2020

- Conducted and moderated weekly code review sessions for CS 126
- Gave students constructive criticism to facilitate programming skill growth
- Graded CS 374 assignments and gave students feedback

PintOS

June 2020 — August 2020

- Implemented kernel space exec, file operations, memory operations system call functions and memory subsystem
- Designed a priority scheduler with priority inheritance protocol
- Designed an inode-based filesystem with memory caching of recently used blocks
- Implemented a shell for user space operations in PintOS

SKILLS

Programming Languages: C, C++, x86 Assembly, MIPS Assembly, RISC-V Assembly, Python, Haskell, Java, JavaScript

Software & Technology: Bash, Git, Nginx, PyTorch, JUnit, Catch2, GoogleTest, $\LaTeX$

Operating Systems: GNU/Linux, Windows, Mac OS